

Lecture 7

Microeconomic Theory 3 – ECON 709

Fall 2026

Recap' Lecture 6: Choice

- ▶ A choice rule C is a correspondence which assigns to every choice problem $A \subseteq X$ in D (the collection of all relevant choice problems, i.e. the domain of C) a non-empty subset of acceptable choices $\emptyset \neq C(A) \subseteq A$.
- ▶ Given a preference relation $\succsim$, the **induced choice rule** $C_{\succsim}$ is defined by $C_{\succsim}(A) = \{x \in A \mid x \succsim y \text{ for all } y \in A\}$.
- ▶ A preference relation $\succsim$ **rationalizes** a choice rule C if $C = C_{\succsim}$ on the domain of C and $\succsim$ is rational.
- ▶ Given a choice rule C , its **revealed preference relation** $\succsim_C$ is defined by $x \succsim_C y$ if and only if $x, y \in A$ and $x \in C(A)$ for some $A \in D$.
- ▶ The choice rule C satisfies the **weak axiom of revealed preference** if the following property holds:
If $x, y \in A \cap B$ and $x \in C(A)$ and $y \in C(B)$, then $x \in C(B)$.

- ▶ If a choice rule C satisfies the WA, then it satisfies Sen's α :
If $x \in A \subseteq B$ and $x \in C(B)$, then $x \in C(A)$.
- ▶ If a choice rule C satisfies the WA, then it satisfies Sen's β :
If $x, y \in A \subseteq B$, $x, y \in C(A)$ and $y \in C(B)$, then $x \in C(B)$.
- ▶ Suppose that $\succsim$ is a rational preference relation. Then the induced choice rule $C_{\succsim}$ satisfies the weak axiom.
 $\Rightarrow$ A choice rule has to satisfy WA in order to be rationalizable.
- ▶ What is a sufficient condition for rationalizability?

WA is sufficient for rationalizability only with additional assumptions on D .

Example:

- ▶ Suppose that $X = \{x, y, z\}$, $D = \{\{x, y\}, \{y, z\}, \{x, z\}\}$ and $C(\{x, y\}) = \{x\}$, $C(\{y, z\}) = \{y\}$ and $C(\{x, z\}) = \{z\}$.
- ▶ C trivially satisfies WA because there are no distinct $x, y \in X$ and $A, B \in D$ such that $x, y \in A \cap B$.
- ▶ There is no preference relation which rationalizes C :
 $C(\{x, y\}) = \{x\}$ and $C(\{y, z\}) = \{y\}$ imply that it would be necessary to have $x \succ y$ and $y \succ z$. Transitivity of $\succsim$ would then require that $x \succ z$ but this contradicts $C(\{x, z\}) = \{z\}$.

Intuition:

- ▶ The more subsets of X are contained in D , the more WA restricts a choice rule C
and if D includes enough subsets of X , then the restrictions of WA on C are strong enough for C to be rationalizable.
- ▶ With too few subsets of X in D , WA might not rule out choice behavior which is incompatible with a rational preference relation.
- ▶ E.g. in above example, including X in D would make a choice rule which satisfies WA rationalizable.

WA as a sufficient condition for rationalizability

Proposition 11

Assume that C is a choice rule which satisfies the weak axiom and that D includes at least all subsets of X of size at most 3. Then C is rationalizable, that is, there exists a rational preference relation $\succsim$ such that $C = C_{\succsim}$ on the domain of C . Moreover, $\succsim$ is the unique preference relation which rationalizes C .

Proof.

A natural candidate for a rationalizing preference relation is the revealed preference relation $\succsim_C$. We show first that $\succsim_C$ is rational, second that $C = C_{\succsim_C}$ on the domain of C , and third that no other preference relation rationalizes C .

1. Reflexivity and completeness: Since C is non-empty, for all $x, y \in X$, $x \in C(\{x, y\})$ or $y \in C(\{x, y\})$. Thus $x \succsim_C x$ for all $x \in X$, and $x \succsim_C y$ or $y \succsim_C x$ for any distinct $x, y \in X$.

Transitivity: Let $x \succsim_C y$ and $y \succsim_C z$. Since $C(\{x, y, z\}) \neq \emptyset$, x, y , or z in $C(\{x, y, z\})$. If $y \in C(\{x, y, z\})$, then $x \in C(\{x, y, z\})$ by the WA. If $z \in C(\{x, y, z\})$, then $y \in C(\{x, y, z\})$ by the WA and hence $x \in C(\{x, y, z\})$. Thus in any case $x \in C(\{x, y, z\})$ and $x \succsim_C z$ by the definition of $\succsim_C$.

2. Suppose $x \in C(B)$. Then $x \succsim_C y$ for all $y \in B$ and thus $x \in C_{\succsim_C}(B)$.

Suppose $x \in C_{\succsim_C}(B)$. This implies $x \succsim_C y$ for all $y \in B$. Since $C(B) \neq \emptyset$, there is $y \in C(B)$. If $y \neq x$, then $x \in C(B)$ because y cannot be revealed strictly better than x by WA. Thus, $x \in C_{\succsim_C}(B)$ if and only if $x \in C(B)$, that is, $C = C_{\succsim_C}$.

3. D includes all two-element subsets of X . Hence the choice behavior in C completely determines the pairwise preference relations over X of a rationalizing preference relation; no other than the revealed preference relation can be rationalizing. $\square$

So far we have discussed preferences and choice in an abstract setting. Now, we are going to move to a special case of an economic agent in more detail: preferences and choice of a consumer.

Consumer Preferences

based on: Rubinstein, lecture 4; MWG, sections 2.A-2.C, 3.A-3.C

The consumer's world

- ▶ the consumer lives in a market economy with commodities (goods and services) that can be obtained in exchange for money at known prices (or equivalently, in exchange for other commodities at known exchange rates)
- ▶ the consumer makes choices between different available combinations of commodities
- ▶ K commodities
- ▶ a *bundle* $x = (x_1, \dots, x_K)$ is a combination of the K commodities where x_k is the quantity of commodity k
- ▶ the set of all available alternatives, X , is the *consumption set*: the set of all bundles which can potentially be consumed given the physical constraints of the environment
- ▶ assume that $X = \mathbb{R}_+^K = \{x = (x_1, \dots, x_K) \mid \text{for all } k, x_k \geq 0\}$ (all non-negative bundles of commodities), unless explicitly stated otherwise

Consumer's preferences over X

- ▶ from now on, we assume that a consumer's preference relation $\succsim$ over X is rational
- ▶ we will discuss additional properties of $\succsim$ in the following, which rely on the particular nature of X :
 - ▶ monotonicity: relies on the ability to compare bundles according to the amount of any particular commodity
 - ▶ convexity: relies on the ability to speak of the sum of two bundles and the multiplication of a bundle by a scalar
- ▶ we will look at special classes of preference relations: homothetic, quasi-linear
- ▶ we will often demonstrate properties of the consumer's preferences by referring to the map of indifference curves, where an indifference curve is a set of the type $\{y | y \sim x\}$ for some bundle x

Monotonicity: "more is better"

Definition 12

The preference relation $\succsim$ on X is *monotone* if for all $x, y \in X$, $x_k \geq y_k$ for all k implies $x \succsim y$, and $x_k > y_k$ for all k implies $x \succ y$.

The preference relation $\succsim$ on X is *strongly monotone* if for all $x, y \in X$, $x_k \geq y_k$ for all k and $x \neq y$ implies $x \succ y$.

- Strong monotonicity obviously implies monotonicity.

- ▶ Monotonicity allows for indifference between bundles in which some quantities are equal. E.g. if $x_1 > y_1$ and $x_2 = y_2$, then $x \succsim y$ and $y \succsim x$ as well is not ruled out. Thus we can have $(x_1, x_2) \sim (y_1, y_2)$. This would not be possible with strong monotonicity.
- ▶ Our definition of monotonicity is a stronger version of MWG, who do not include the requirement that if $x_k \geq y_k$ for all k , then $x \succsim y$. But of course if preferences are monotone according to our definition, then they are monotone according to MWG.

Monotonicity and Utility Representations

- ▶ A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is strictly increasing if $x_k \geq y_k$ for all k implies $f(x) \geq f(y)$, and if $x_k > y_k$ for all k implies $f(x) > f(y)$.

A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is strongly increasing if $x_k \geq y_k$ for all k and $x \neq y$ implies $f(x) > f(y)$.

The following result is easy to check:

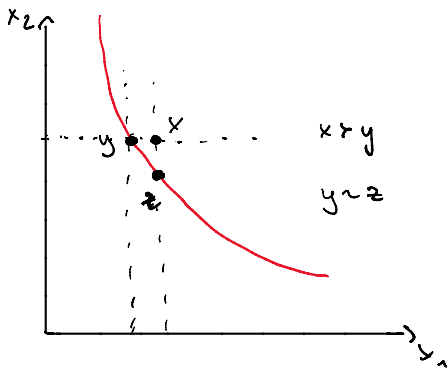
- ▶ If u represents preference relation $\succsim$,
 - ▶ $\succsim$ is monotone if and only if u is strictly increasing.
 - ▶ $\succsim$ is strongly monotone if and only if u is strongly increasing.

Examples of (strongly) monotone preference relations

- ▶ If a preference relation $\succsim$ on $X = \mathbb{R}_+^2$ is represented by $u(x) = \min \{x_1, x_2\}$, then $\succsim$ is monotone, but not strongly monotonic.
- ▶ If a preference relation $\succsim$ on $X = \mathbb{R}_+^2$ is represented by $u(x) = x_1 + x_2$, then $\succsim$ strongly monotone.

Strong monotonicity implies strictly decreasing indifference curves

Let $K = 2$. If $y \sim z$ and $y_1 < z_1$, then $z_2 < y_2$ by strong monotonicity. If $z_2 \geq y_2$, then strong monotonicity would imply that $z \succ y$, a contradiction.



Local Non-Satiation

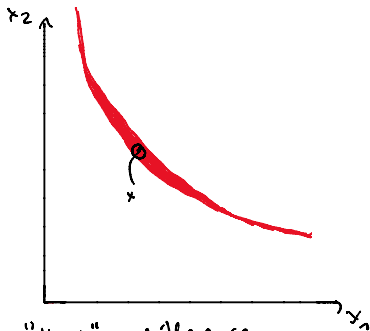
A weaker "more is better" property than monotonicity:

Definition 13

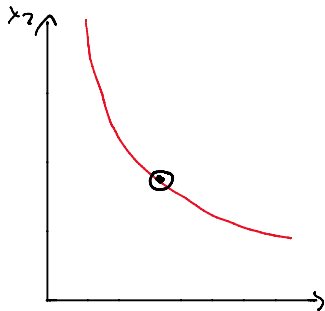
The preference relation $\succsim$ is *locally non-satiated* if for every $x \in X$ and every $\epsilon > 0$, there exists $y \in X$ such that $\|y - x\| \leq \epsilon$ and $y \succ x$.

$\|x - y\|$ is the Euclidean distance between points x and y .

- ▶ Local non-satiation says that for any x and any arbitrarily small distance away from x , denoted by $\epsilon > 0$, there is another bundle y within this distance from x that is strictly preferred to x .
- ▶ A preference relation with "thick" indifference sets is not locally non-satiated.



"thick" indifference curves violate local non-satiability.



local non-satiability x_1 implies "thin" indifference curves

Relation between Local Non-Satiation and Monotonicity

- ▶ If a preference relation is monotonic, then it is also non-satiated:

For any x and any $\epsilon > 0$, y with $y_k \in (x_k, x_k + \epsilon)$ for all k is strictly preferred to x by monotonicity.

- ▶ But not every locally non-satiated preference relation is monotonic:

Suppose $X = \mathbb{R}_+^2$ and let $x \succsim y$ if and only if $|x_1 - x_2| \geq |y_1 - y_2|$. This is a preference relation which is locally non-satiated and not monotonic.

Convexity: "Moderation is at least as good as the worse of two extremes"

Convex preferences capture two related intuitions in situations where we can speak of an "in between" two alternatives:

- ▶ If x is weakly preferred over y , then any move from y to x is also a weak improvement over y :
Suppose you prefer living in the center of Montreal over living in the suburbs, then any move closer to the center of Montreal is better than living in the suburb.
- ▶ If z is between x and y , then it is impossible that both x and y are better than z :
Consider preferences over political candidates. Any candidate between the most left and the most right is weakly preferred over the worse of the two extreme candidates.

Does convexity always make sense? Consider convex combinations of milk and orange juice...

Definition of Convexity

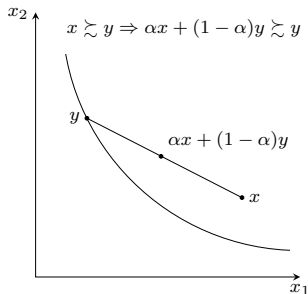
We will see now 3 different definitions of convexity each of which reveals a different intuition but which are all equivalent.

Thus, if we say a preference relation is convex, you can have any of the following three definitions in mind.

3 Equivalent Convexity Definitions

Definition 14 (Convexity 1.)

The preference relation $\succsim$ satisfies convexity 1, if $x \succsim y$ implies $\alpha x + (1 - \alpha)y \succsim y$ for all $\alpha \in [0, 1]$.



Convexity 1

If x is weakly preferred over y , then any move from y to x is also a weak improvement over y .

Definition 15 (Convexity 2.)

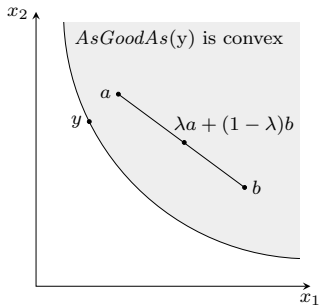
The preference relation $\succsim$ satisfies convexity 2, if for all x, y and $z = \alpha x + (1 - \alpha)y$ with $\alpha \in [0, 1]$, $z \succsim x$ or $z \succsim y$.

If z is between x and y , then it is impossible that both x and y are better than z ;

z is at least as good as the worse alternative among x and y .

Definition 16 (Convexity 3.)

The preference relation $\succsim$ satisfies convexity 3, if for all $x \in X$, the upper contour set $\{y | y \succsim x\}$ is convex; that is, if $y \succsim x$ and $z \succsim x$, then $\alpha y + (1 - \alpha)z \succsim x$ for any $\alpha \in [0, 1]$.



Convexity 3

C3 captures the intuition that if both z_1 and z_2 are better than y , then an average of z_1 and z_2 is also preferred to y .

Equivalence Proof

Proposition 12

If the preference relation $\succsim$ satisfies one of the conditions convexity 1, convexity 2, and convexity 3, then it satisfies the other two as well.

Proof.

$C1 \Rightarrow C2$: Assume that $\succsim$ satisfies convexity 1 and let $x, y, z \in X$ such that $z = \alpha x + (1 - \alpha)y$ for some $\alpha \in (0, 1)$. Without loss of generality, assume that $x \succsim y$. By convexity 1, we have $z \succsim y$. Thus, $\succsim$ satisfies convexity 2.

$C2 \Rightarrow C3$: Assume that $\succsim$ satisfies convexity 2 and let $y \succsim x$ and $z \succsim x$. Then, by convexity 2, $\alpha y + (1 - \alpha)z$ is at least as good as either y or z (or both). In any case, by transitivity, $\alpha y + (1 - \alpha)z \succsim x$, and thus $\succsim$ satisfies convexity 3.

$C3 \Rightarrow C1$: Assume that $\succsim$ satisfies convexity 3. If $x \succsim y$, then both $x \succsim y$ and $y \succsim y$ and thus $\alpha x + (1 - \alpha)y \succsim y$. Thus, $\succsim$ satisfies convexity 1.

Strict Convexity

Stronger version of convexity:

"An average is **strictly** preferred to the worse one of the two extremes."

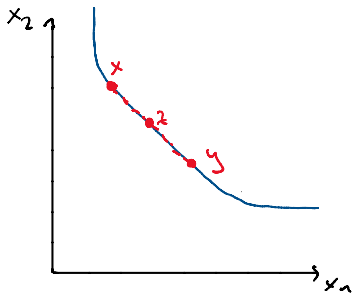
Definition 17

*The preference relation $\succsim$ is **strictly convex**, if $y \succsim x$, $z \succsim x$, and $y \neq z$ imply $\alpha y + (1 - \alpha)z \succ x$ for all $\alpha \in (0, 1)$.*

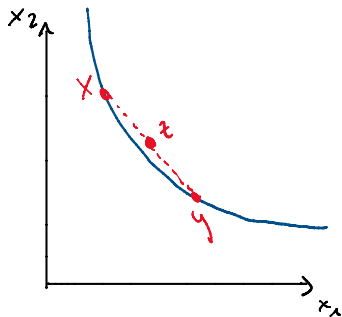
Above is the strict convexity analogue of C3.

Strict convexity analogue of definitions C1 and C2: strict preference implication.

Still equivalence between strict convexity analogues of definitions C1-C3.



- $x \succeq y$, $z \succeq y$ and not $z \succeq x$.
- convex, but not strictly convex prefs.



- $x \succeq y$ and $z \succ y$,
- strictly convex prefs.

Example of Convex Preferences

Let $X = \mathbb{R}_+^2$ and the preference relation $\succsim$ be defined by $x \succsim y$ if and only if $x_1 + x_2 \geq y_1 + y_2$.

- ▶ $\succsim$ is convex:

Let $x \succsim y$, i.e. $x_1 + x_2 \geq y_1 + y_2$. Then

$\alpha x + (1 - \alpha)y = (\alpha x_1 + (1 - \alpha)y_1, \alpha x_2 + (1 - \alpha)y_2)$ for $\alpha \in (0, 1)$. Now

$$\begin{aligned} & [\alpha x_1 + (1 - \alpha)y_1] + [\alpha x_2 + (1 - \alpha)y_2] \\ &= \alpha(x_1 + x_2) + (1 - \alpha)[y_1 + y_2] \\ &\geq \alpha(y_1 + y_2) + (1 - \alpha)[y_1 + y_2] = y_1 + y_2. \end{aligned}$$

Thus $\alpha x + (1 - \alpha)y \succsim y$.

- ▶ $\succsim$ is not strictly convex:

Consider bundles $(1, 0)$ and $(0, 1)$. We know that

$(1, 0) \succsim (0, 1)$ and $(0, 1) \succsim (0, 1)$, and $(1, 0) \neq (0, 1)$. Take $\frac{1}{2}(1, 0) + \frac{1}{2}(0, 1) = (\frac{1}{2}, \frac{1}{2})$. However $(\frac{1}{2}, \frac{1}{2}) \sim (0, 1)$ and not $(\frac{1}{2}, \frac{1}{2}) \succ (0, 1)$.

Utility Representations of Convex Preferences

- ▶ A function $u : X \rightarrow \mathbb{R}$ is concave if for all x, y , and $\lambda \in [0, 1]$ we have $u(\lambda x + (1 - \lambda)y) \geq \lambda u(x) + (1 - \lambda)u(y)$, and is quasi-concave if for all y , the set $\{x | u(x) \geq u(y)\}$ is convex, or equivalently, if $u(\lambda x + (1 - \lambda)y) \geq \min \{u(x), u(y)\}$ for all x, y and $\lambda \in [0, 1]$.
- ▶ Trivially, any concave function is also quasi-concave.

Proposition 13

If a preference relation $\succsim$ is represented by a utility function, then $\succsim$ is convex if and only if u is quasi-concave.

Proof left as exercise.

Thus, a concave utility function implies convex preferences.

However, convexity of a preference relation does not imply that its utility representation is concave!